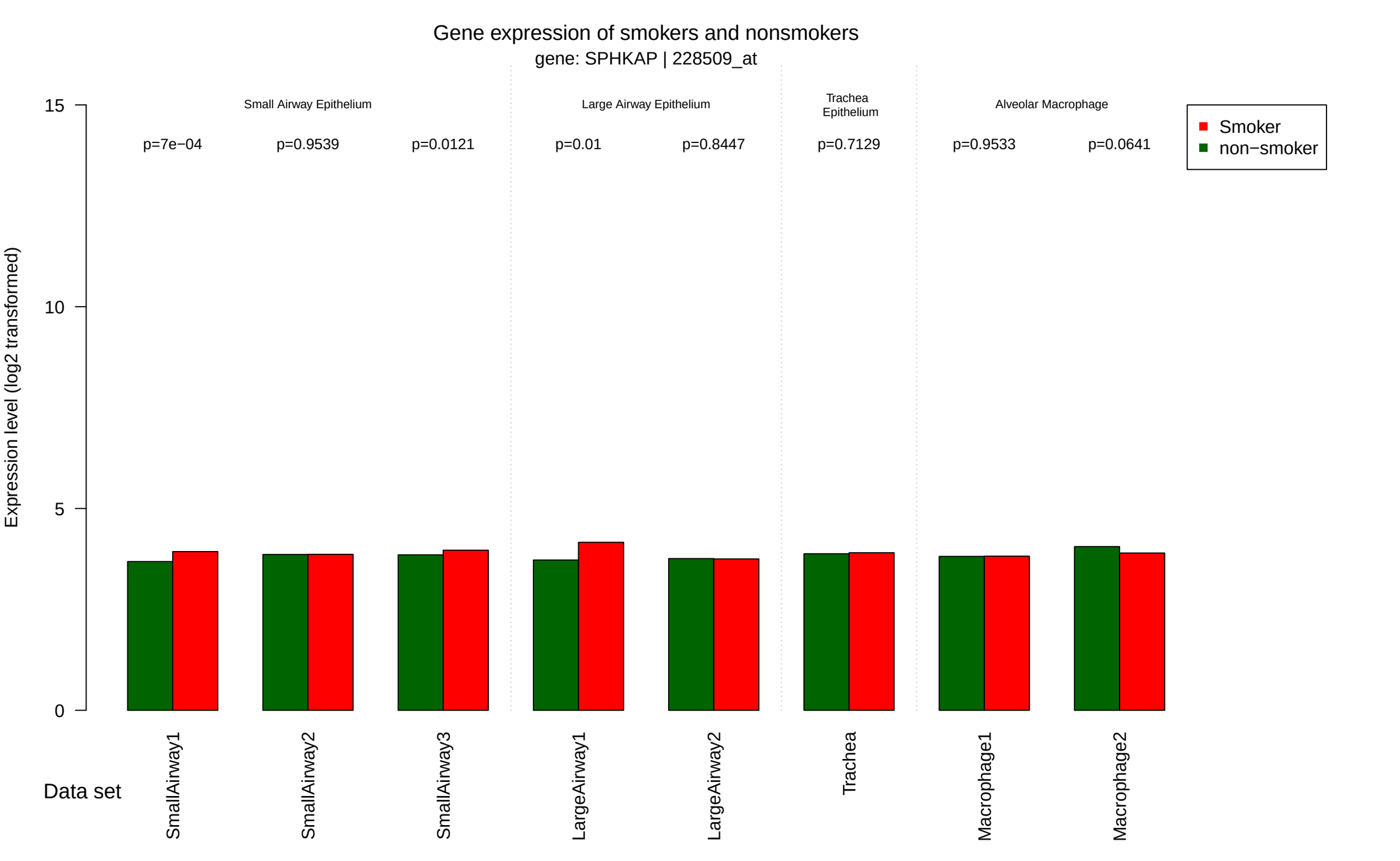
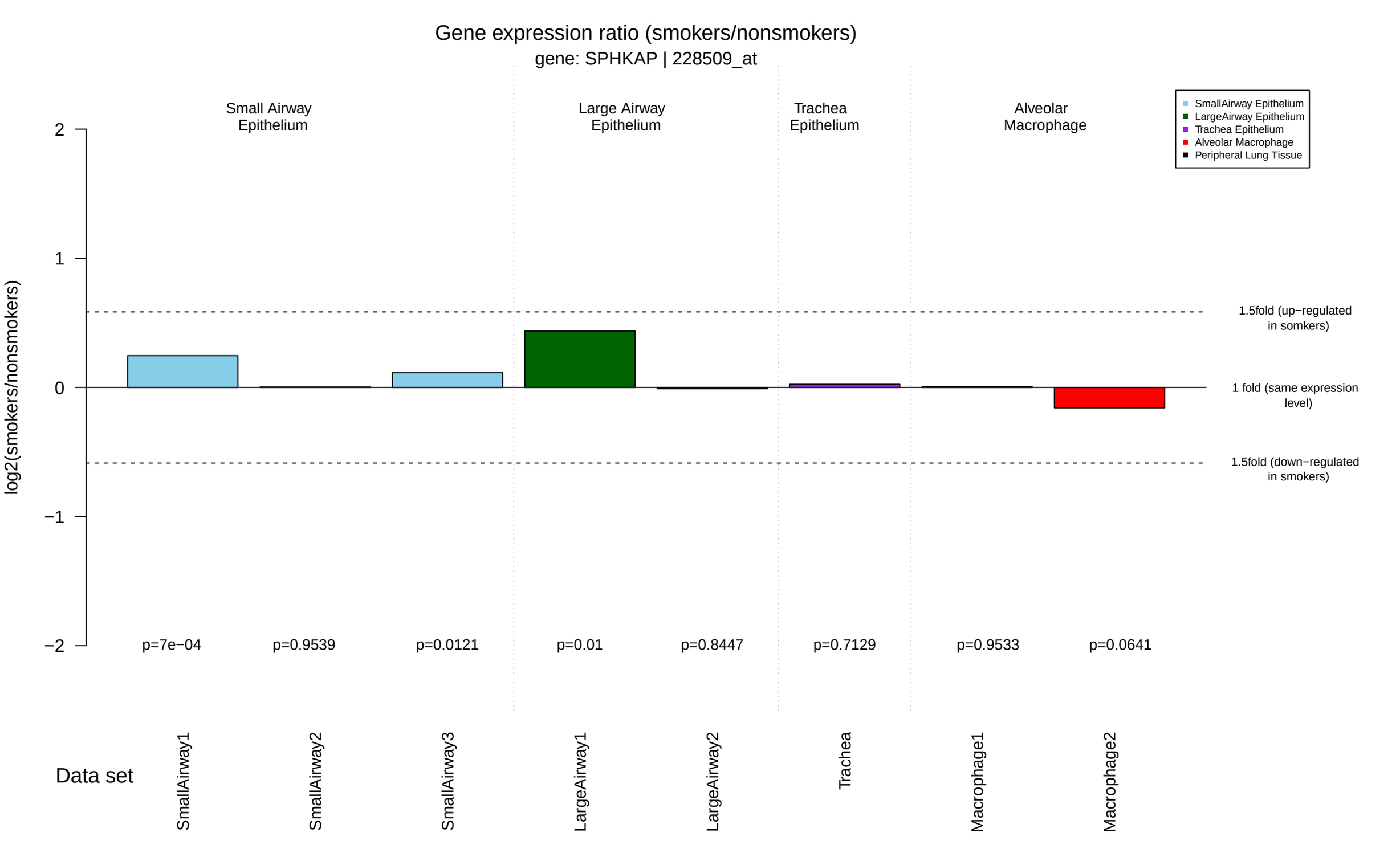
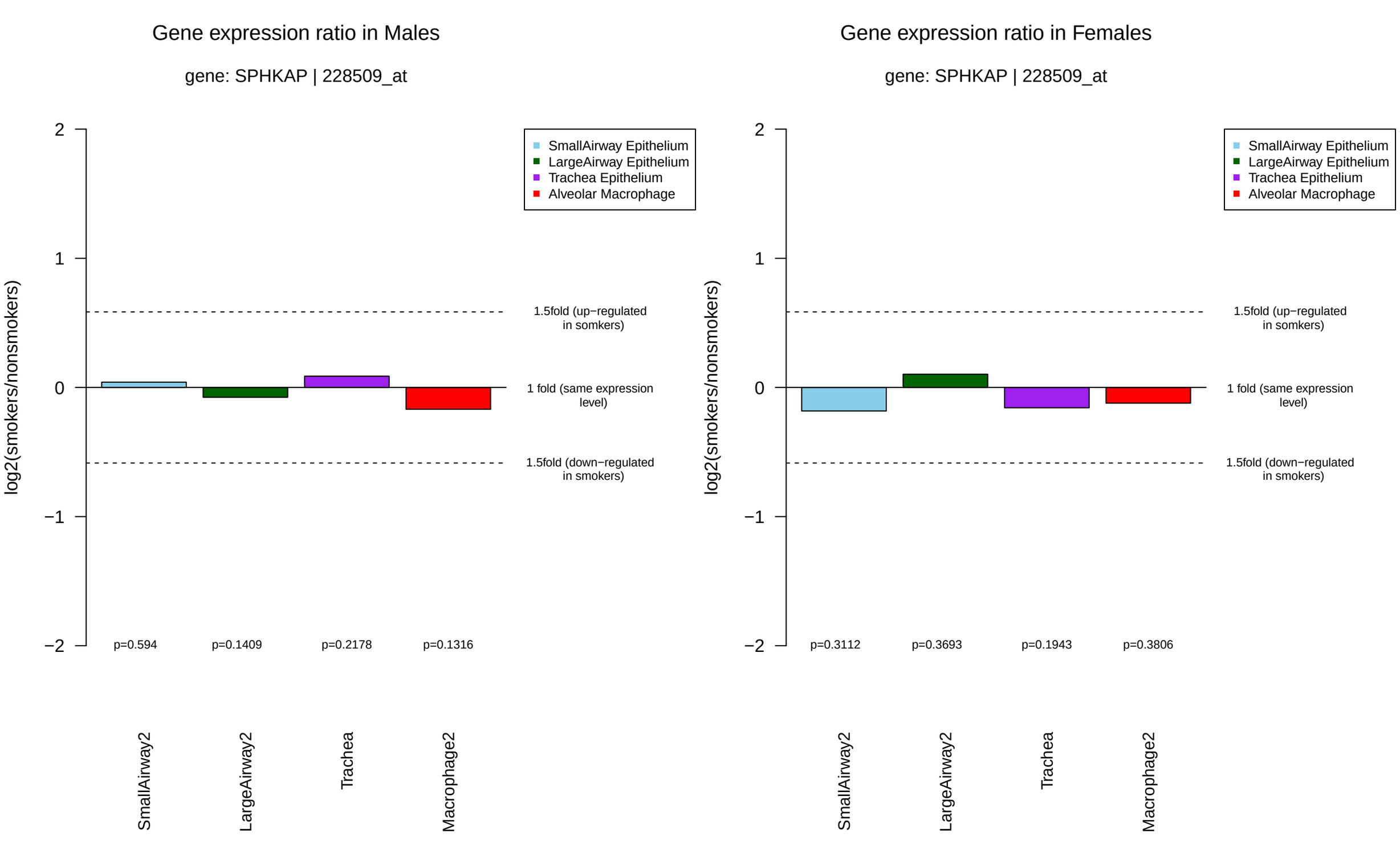
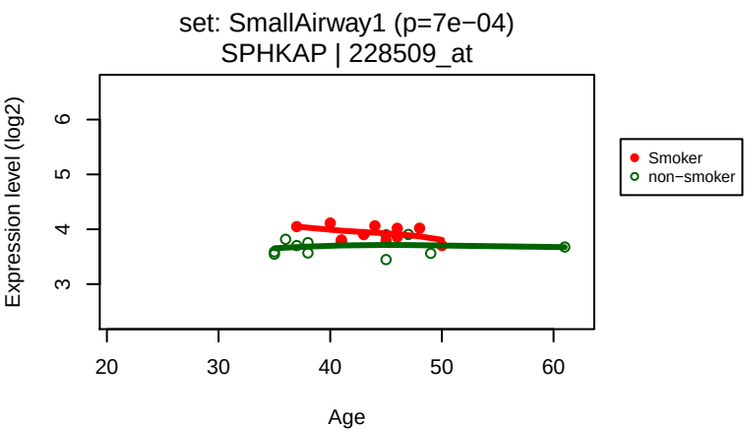
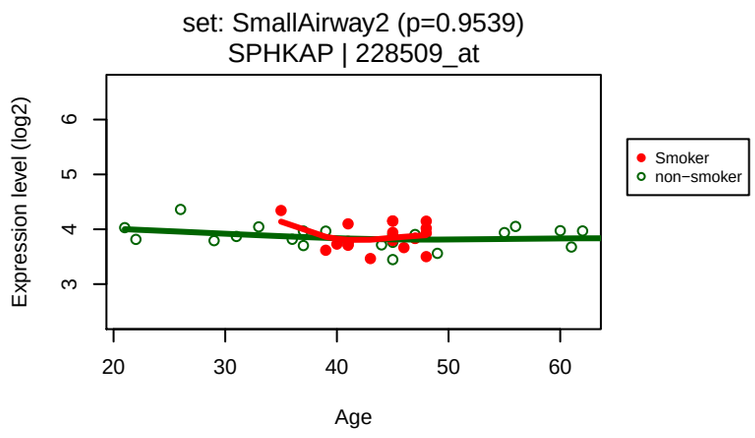
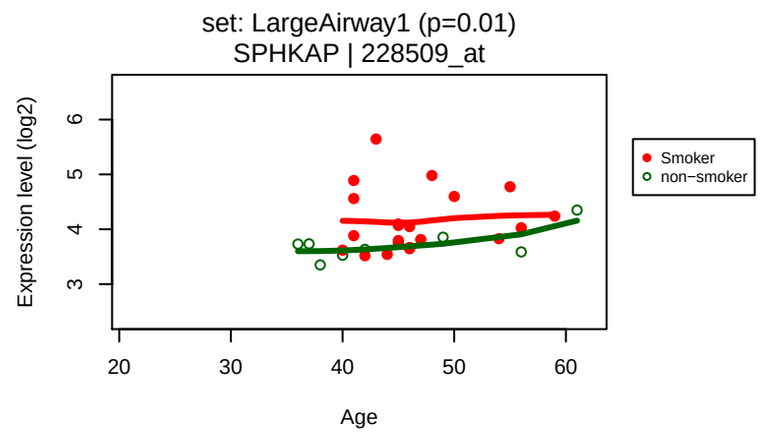

Gene expression of smokers and nonsmokers
gene: SPHKAP | 228509_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=7e−04
p=0.9539
p=0.0121
p=0.01
p=0.8447
p=0.7129
p=0.9533
p=0.0641
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

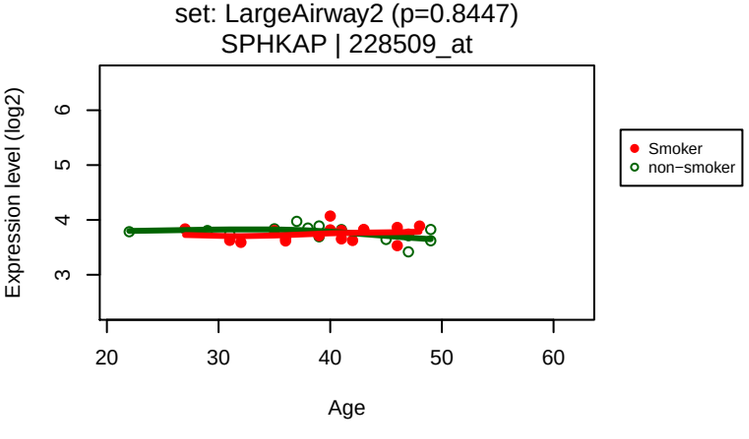

Gene expression ratio (smokers/nonsmokers)
gene: SPHKAP | 228509_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=7e−04
p=0.9539
p=0.0121
p=0.01
p=0.8447
p=0.7129
p=0.9533
p=0.0641
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

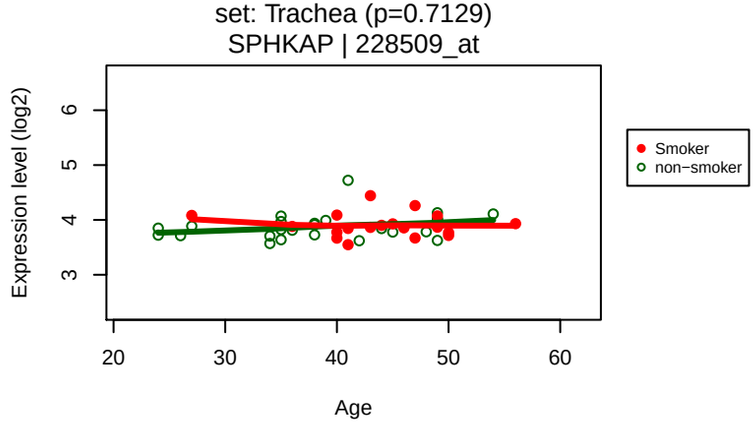

Gene expression ratio in Males
gene: SPHKAP | 228509_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.594
p=0.1409
p=0.2178
p=0.1316
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPHKAP | 228509_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3112
p=0.3693
p=0.1943
p=0.3806
SmallAirway2
LargeAirway2
Trachea
Macrophage2

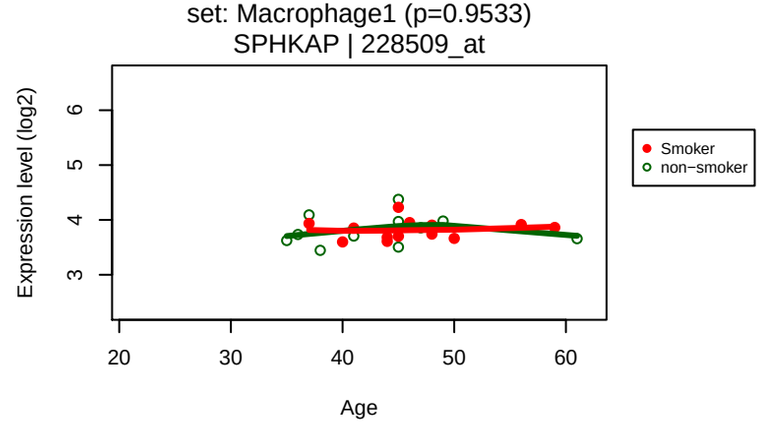

set: SmallAirway1 (p=7e−04)
SPHKAP | 228509_at
Expression level (log2)
Age
Smoker
non−smoker

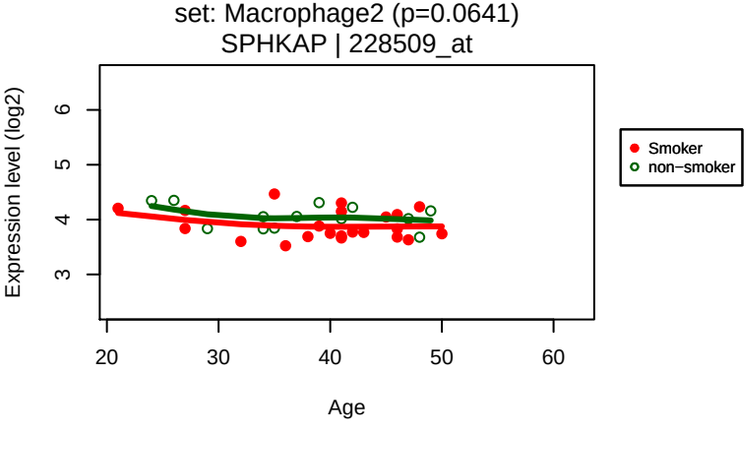

set: SmallAirway2 (p=0.9539)
SPHKAP | 228509_at
Expression level (log2)
Age
Smoker
non−smoker

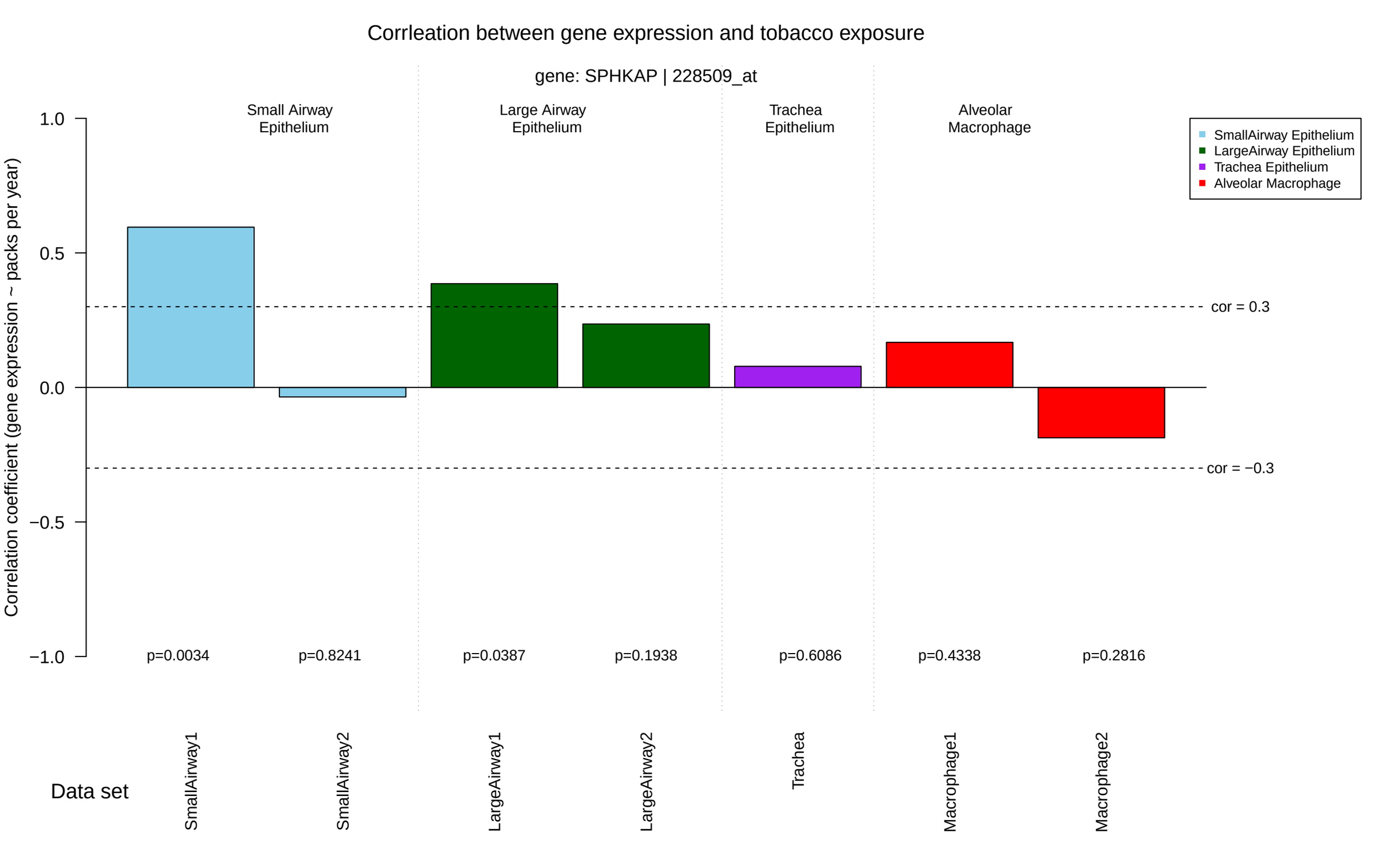

set: LargeAirway1 (p=0.01)
SPHKAP | 228509_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8447)
SPHKAP | 228509_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7129)
SPHKAP | 228509_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9533)
SPHKAP | 228509_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0641)
SPHKAP | 228509_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPHKAP | 228509_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0034
p=0.8241
p=0.0387
p=0.1938
p=0.6086
p=0.4338
p=0.2816
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2